

SONICGEN

Audio-Driven Player DNA & Dynamic Soundtrack AI

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The Problem Space

Modern video games utilize **static soundtracks** that fail to adapt to individual player behaviors, emotional states, or real-time tactical decisions.

Furthermore, current player profiling relies heavily on rigid telemetry data:

- Kill/Death Ratios
- Movement Speed & Heatmaps
- Objective Completion Rates

The Gap: These metrics ignore the most direct indicator of player intent and state—**how they communicate** via voice chat.

The Solution & Business Value

An AI system that ingests a player's voice communications, extracts an **8-Axis Gamer DNA profile**, and dynamically generates custom background music to match their exact playstyle.

Value Proposition:

- **Enhanced Immersion:** Players remain in a "flow state" longer when auditory feedback matches their emotional intensity.
- **Player Retention:** Hyper-personalized gaming experiences increase overall Lifetime Value (LTV) and session length.
- **Esports & Streaming:** Content creators automatically receive copyright-free, adaptive background music tailored to their gameplay intensity.

Dataset & Inputs

To evaluate the pipeline, we utilized simulated voice communication data designed to mimic real-world gaming environments.

- **Speech Input:** 10-second simulated audio clips of player voice chat over standard comms channels.
- **Scenario A (Aggressive):** "Push mid, they're weak! Flank them right now!"
- **Scenario B (Tactical/Calm):** "Let's just gather resources and build a base quietly. Watch the perimeter."

Target Output:

10-second custom AI-generated soundtracks mapped to the deduced behavioral DNA.

Pipeline Architecture

Phase 1: Speech-to-Text

Using `openai/whisper-small` to accurately transcribe the raw player voice chat, handling varying speeds and intensities.

Phase 2: DNA Extraction (Agentic Logic)

An NLP logic layer analyzes the transcription to determine the "8-Axis Gamer DNA" (e.g., Aggression, Tactical, Stealth, Chaos) and synthesizes a highly descriptive music prompt.

Phase 3: Music Generation

Using `facebook/musicgen-small` to take the synthesized prompt and render a seamless, customized soundtrack on the fly.

Prompt & Input Engineering

Effective generation requires translating abstract player states into concrete audio engineering parameters.

High Aggression DNA Mapping:

Prompt: "Heavy bass, fast tempo 140bpm, aggressive synth, cyberpunk, high energy battle sequence."

High Stealth / Calm DNA Mapping:

Prompt: "Dark ambient, slow tempo, tense strings, quiet atmospheric thriller music, sparse percussion."

Evaluation Methodology

Baseline vs. Improved Pipeline

- **Baseline:** Feeding MusicGen generic static prompts (e.g., "video game combat music").
- **Improved:** Injecting the dynamically extracted 8-Axis DNA prompt structure mapped to BPM and specific instrument clusters.

Key Metrics Tracked

- **Whisper WER (Word Error Rate):** Evaluated transcription accuracy on gaming-specific slang and rapid speech.
- **Music Prompt Alignment:** Qualitative evaluation of whether the generated audio matched the detected emotional state of the voice chat.

Results & Insights

- **Finding 1:** Whisper struggles slightly with highly niche gaming slang without specific fine-tuning, but it perfectly captures the **general intent and action verbs** necessary for DNA profiling.
- **Finding 2:** MusicGen responds remarkably well to explicit BPM (Beats Per Minute) and genre tags triggered by the Agentic DNA logic, producing reliable stylistic shifts.
- **Conclusion:** Multimodal chaining (Speech → Text → Music) successfully circumvents static game design, allowing for highly coherent, real-time audio personalization.

Limitations & Future Work

Current Limitations:

- **Compute Latency:** MusicGen requires several seconds to generate audio segments. This currently limits instantaneous, split-second in-game adaptation without pre-buffering.

Future Expansions:

- **Granular Sound Effects:** Integrating AudioLDM2 to generate specific, one-off sound effects (e.g., customized weapon sounds or alert pings) based on the same continuous Player DNA profile.
- **Continuous Profiling:** Implementing graph machine learning to map player DNA evolution over long campaign playtimes.